

# Deep Learning — Autumn 2026

## Course Syllabus & Lecture Schedule

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**Audience:** Master's & Ph.D. students | **Format:** 20 lectures × 2 h = 40 contact hours | **Schedule:** Sat & Sun, 1 Aug – 4 Oct 2026

| #                                                          | Date        | Day | Lecture Topic                  | Key Contents                                                                                                           |
|------------------------------------------------------------|-------------|-----|--------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>MODULE 1 — INTRODUCTION</b>                             |             |     |                                |                                                                                                                        |
| 1                                                          | 01 Aug 2026 | Sat | Introduction to Deep Learning  | History (McCulloch–Pitts → DL revolution), representation learning, learning paradigms, model families, course roadmap |
| <b>MODULE 2 — MATHEMATICAL FOUNDATIONS</b>                 |             |     |                                |                                                                                                                        |
| 2                                                          | 02 Aug 2026 | Sun | Math Foundations I             | Linear algebra, eigen-decomposition, SVD/PCA, gradients, Jacobian, Hessian, entropy, cross-entropy, KL, MI             |
| 3                                                          | 08 Aug 2026 | Sat | Math Foundations II            | Probability, distributions, MLE/MAP, bias–variance, random processes, sampling, convolution, DFT/FFT                   |
| <b>MODULE 3 — NEURAL NETWORKS &amp; LEARNING</b>           |             |     |                                |                                                                                                                        |
| 4                                                          | 09 Aug 2026 | Sun | Neural Networks I              | Perceptron & convergence, XOR, MLPs, activation functions, universal approximation, width vs. depth                    |
| 5                                                          | 15 Aug 2026 | Sat | Neural Networks II             | Hypothesis classes, ERM, generalization, loss functions, computational graphs, backpropagation, autodiff               |
| <b>MODULE 4 — OPTIMIZATION &amp; REGULARIZATION</b>        |             |     |                                |                                                                                                                        |
| 6                                                          | 16 Aug 2026 | Sun | Optimization & Regularization  | SGD, momentum, Nesterov, AdaGrad, RMSProp, Adam(W), loss landscapes, Newton, dropout, BatchNorm/LayerNorm/RMSNorm      |
| <b>MODULE 5 — CONVOLUTIONAL NEURAL NETWORKS</b>            |             |     |                                |                                                                                                                        |
| 7                                                          | 22 Aug 2026 | Sat | CNN I: Fundamentals            | Local connectivity, weight sharing, 1D/2D convolution, stride, padding, dilation, receptive fields, pooling            |
| 8                                                          | 23 Aug 2026 | Sun | CNN II: Architectures          | LeNet, AlexNet, ZFNet, VGG, Inception, ResNet, depthwise/separable & dilated convolutions                              |
| 9                                                          | 29 Aug 2026 | Sat | CNN III: Visual Understanding  | Transfer learning, R-CNN family, segmentation, saliency & visualization, adversarial examples, ViT                     |
| <b>MODULE 6 — SEQUENTIAL MODELS</b>                        |             |     |                                |                                                                                                                        |
| 10                                                         | 30 Aug 2026 | Sun | RNN I: RNNs & BPTT             | Recurrence, unrolling, BPTT derivation, vanishing/exploding gradients, truncated BPTT, sequence tasks                  |
| 11                                                         | 05 Sep 2026 | Sat | RNN II: Embeddings, LSTM, GRU  | One-hot vs. distributed representations, Word2Vec (CBOW, skip-gram, neg. sampling), LSTM, GRU gating                   |
| 12                                                         | 06 Sep 2026 | Sun | Seq2Seq, CTC & Beam Search     | Encoder–decoder, teacher forcing, exposure bias, CTC loss & forward–backward, greedy/beam decoding                     |
| <b>MODULE 7 — ATTENTION, TRANSFORMERS &amp; LLMs</b>       |             |     |                                |                                                                                                                        |
| 13                                                         | 12 Sep 2026 | Sat | Attention Mechanisms           | Query–key–value, scaled dot-product, self-/cross-attention, multi-head attention, dynamic routing                      |
| 14                                                         | 13 Sep 2026 | Sun | Transformers                   | Full architecture, residuals & LayerNorm, causal masking, sinusoidal/relative/RoPE, BERT, GPT, T5                      |
| 15                                                         | 19 Sep 2026 | Sat | Large Language Models          | Tokenization, pretraining, scaling laws, KV cache, SFT, LoRA, RLHF, DPO, decoding, RAG, open challenges                |
| <b>MODULE 8 — GENERATIVE DEEP LEARNING</b>                 |             |     |                                |                                                                                                                        |
| 16                                                         | 20 Sep 2026 | Sun | Autoencoders                   | Latent representations; undercomplete, sparse, denoising, contractive AEs; link to PCA; applications                   |
| 17                                                         | 26 Sep 2026 | Sat | Variational Autoencoders       | Latent-variable models, amortized inference, reparameterization, ELBO derivation, $\beta$ -VAE, CVAE                   |
| 18                                                         | 27 Sep 2026 | Sun | Flows, Diffusion & GANs        | Change of variables, RealNVP/Glow, DDPM, score matching, latent diffusion, CFG, GAN/WGAN/cGAN                          |
| <b>MODULE 9 — GRAPH, ENERGY &amp; MODERN ARCHITECTURES</b> |             |     |                                |                                                                                                                        |
| 19                                                         | 03 Oct 2026 | Sat | GNNs & Energy-Based Models     | Graph Laplacian, message passing, GCN, GraphSAGE, GAT, MPNN; Hopfield nets, (R)BMs, contrastive divergence             |
| 20                                                         | 04 Oct 2026 | Sun | State-Space Models, S4 & Mamba | Linear SSMS, recurrent vs. convolutional views, S4, selective scan, Mamba; RNN/Transformer/SSM comparison              |

### Primary References

Goodfellow, Bengio & Courville, *Deep Learning* (2016)  
Bishop & Bishop, *Deep Learning: Foundations and Concepts* (2024)  
Murphy, *Probabilistic Machine Learning* (2022–23)  
Original research papers are cited in each lecture.

### Each Lecture Includes

Objectives · motivation · derivations · architectures · pseudocode · worked examples · PyTorch code · key papers · limitations & open problems · discussion questions